B.Tech (Bachelor of Technology) Project 2026-27: Synopsis Presentation

"Multi-Agent RAG (Retrieval-Augmented Generation) System for DPDP (Digital Personal Data Protection) Act Compliance Assessment and Citizen Rights Awareness"

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Project Synopsis Outline

1. **Introduction**
   Background, importance, relevance & purpose of the study

2. **Literature Survey & Need of Work**
   Justification for the project with relevant references

3. **Problem Statement**
   Clear explanation of the issue, its context and aims

4. **Objectives**
   Major & sub-objectives, divided into modules

5. **Proposed Work**
   System architecture, modules, mechanisms & approach

6. **Requirements**
   Hardware, software, tools/platforms & cost

7. **Project Plan**
   Month-wise plan of execution

8. **References**
   IEEE (Institute of Electrical and Electronics Engineers)-format papers and web resources

1. Introduction

Background

India's DPDP (Digital Personal Data Protection) Act, 2023 and DPDP Rules, 2025 make every organisation that handles personal data a Data Fiduciary with legal duties, and give every citizen (Data Principal) enforceable rights.

Importance & Relevance

Duties and penalties of up to ₹250 crore apply from 13 May 2027, yet privacy policies are rarely checked against the Act, and LLMs (Large Language Models) often cite legal sections that do not exist.

Applicability & Purpose

DPDP Compass audits a privacy policy against 14 DPDP obligations with exact citations, drafts compliant policies, and explains citizens' rights in plain English, Hindi and Marathi — one engine for businesses and citizens.

2. Literature Survey and Need of Work

Literature Survey

Sr.	Title	Author/Year	Summary	Limitation
1	PAKTON: Multi-Agent Framework for QA (Question Answering) in Long Legal Agreements	P. Roy et al., 2025	Coordinated agents answer questions over long contracts	QA only; no compliance checking or risk
2	RAGulating Compliance: Multi-Agent KG (Knowledge Graph) for Regulatory QA	B. Agarwal et al., 2025	Extracts triplets from regulations for precise retrieval	Regulation side only; no real policies
3	LegalGraphRAG: Multi-Agent Graph RAG (Retrieval-Augmented Generation)	2026	Organises facts, rules and principles for legal reasoning	Needs a hand-built ontology
4	LLM (Large Language Model) Agents in Law: Taxonomy & Challenges	2026	Survey of LLM-based legal agent systems	No validated system for India's DPDP Act
5	Helping Johnny Make Sense of Privacy Policies with LLMs	2026	Plain-language explanations of privacy policies	Consumer-only; no business audit
6	Large Legal Fictions: Legal Hallucinations in LLMs	M. Dahl et al., 2024	LLMs invent laws, cases and citations	Diagnoses the problem; offers no remedy

2. Literature Survey and Need of Work

Need of the Work

- **No DPDP tool:** no existing system or benchmark is built on India's DPDP (Digital Personal Data Protection) Act.
- **Hallucinated law:** LLMs (Large Language Models) invent sections, so every citation must be verified, not generated.
- **Split tools:** business compliance checkers and citizen-rights explainers are built and evaluated separately.
- **Our difference:** clause-aware retrieval, citations attached by the system, output guardrails, and one rule book that both audits policies and drafts new ones that pass their own audit.
- **Outcome:** one engine serves organisations (Data Fiduciaries) and citizens (Data Principals).

3. Problem Statement

To enable organisations to assess their privacy policies for compliance with the DPDP (Digital Personal Data Protection) Act, 2023, while helping citizens understand their rights and whether those policies uphold them — through a unified AI (Artificial Intelligence)-powered multi-agent RAG (Retrieval-Augmented Generation) system that accurately evaluates policies against DPDP obligations and presents the findings in an accessible manner.

4. Objectives

Primary Objectives

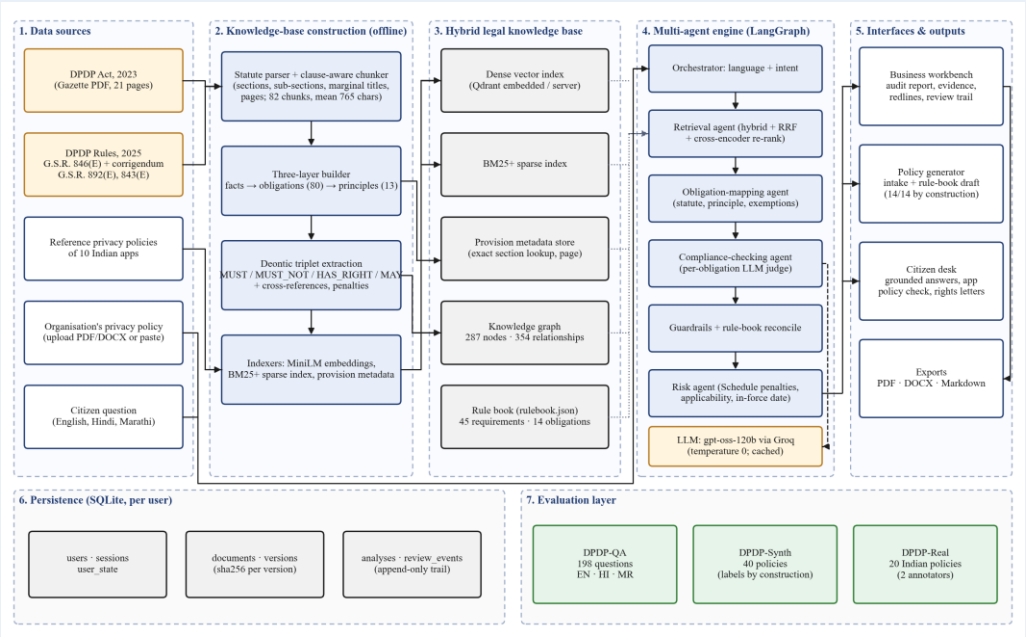
1. Build a multi-agent RAG (Retrieval-Augmented Generation) engine for DPDP Act and Rules 2025 obligations
2. Business module: cited audit report, ranked gaps, fixes and a compliant policy generator
3. Citizen module: plain-language rights answers in English, Hindi and Marathi
4. Evaluate against single-prompt LLM (Large Language Model) and keyword baselines

Sub-Objectives (Module-wise)

- **M1 Knowledge Base:** parse the Gazette PDF (Portable Document Format) into clause chunks and 3 layers
- **M2 Multi-Agent Engine:** hybrid retrieval and 6 LangGraph agents with guardrails
- **M3 Business Compliance:** 14-obligation audit, redlines, rule-book policy generator
- **M4 Citizen Rights:** grounded Q&A (Question and Answer), app policy check, request letters
- **M5 Platform & Evaluation:** accounts, exports, 116 automated tests, benchmarks

5. Proposed Work

System Architecture



5. Proposed Work

Mechanism & Approach

- Clause-aware chunking of the Gazette text
- Hybrid retrieval: dense + BM25+ (Best Matching 25 Plus) + exact section lookup
- RRF (Reciprocal Rank Fusion) + cross-encoder re-ranking
- Per-obligation LLM (Large Language Model) judge
- Guardrails verify every quote and citation
- Rule book: 45 checks; risk from penalties

Project Modules

- **M1** Legal Knowledge Base
- **M2** Multi-Agent Engine
- **M3** Business Compliance (audit, redline, generator)
- **M4** Citizen Rights (Q&A: Question and Answer)
- **M5** Platform & Evaluation

6. Requirements

Hardware Requirements

- Intel i5 or equivalent CPU (Central Processing Unit)
- 8 GB (Gigabyte) RAM (Random Access Memory), 16 GB ideal
- 256 GB SSD (Solid State Drive); stable internet

Software Requirements

- Windows 10/11 or Linux
- Python 3.13, Node.js 20+
- VS (Visual Studio) Code, Git & GitHub, Docker Compose

Devices / Tools / Platforms

- React + TypeScript UI (User Interface)
- FastAPI + LangGraph API (Application Programming Interface)
- Qdrant + SQLite (Structured Query Language) databases
- Groq gpt-oss-120b LLM (Large Language Model)

Cost of Implementation

- Open-source tools and libraries: ₹0
- Self-hosted Qdrant and SQLite: ₹0
- LLM API usage (development & evaluation): ₹3,000–5,000 (estimate)

7. Project Plan

July

- Literature survey
- Gazette + policy data
- Parser and clause-aware chunker

August

- Hybrid retrieval
- Obligation mapping agent
- DPDP-QA (Question Answering) test set

September

- Compliance agent, guardrails, risk
- Business & Citizen UI (User Interface)
- Accounts, versioned documents

October

- Rule book + policy generator
- Evaluation and 116 tests
- Development complete

November

- External examiner review
- Research paper publication
- Final report and demo

8. References

Journal / Conference Papers (IEEE: Institute of Electrical and Electronics Engineers format)

[1] P. Roy et al., “PAKTON: A multi-agent framework for question answering in long legal agreements,” Proc. EMNLP (Empirical Methods in NLP), 2025.

[2] B. Agarwal et al., “RAGulating compliance: A multi-agent knowledge graph for regulatory QA (Question Answering),” arXiv:2508.09893, 2025.

[3] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP (Natural Language Processing) tasks,” NeurIPS (Neural Information Processing Systems), 2020.

[4] M. Dahl et al., “Large legal fictions: Profiling legal hallucinations in LLMs,” J. Legal Analysis, vol. 16, 2024.

[5] “Helping Johnny make sense of privacy policies with LLMs,” arXiv:2501.16033, 2026.

Web Resources (with access dates)

- MeitY (Ministry of Electronics and Information Technology), DPDP Act, 2023, https://www.meity.gov.in, Accessed on: 4 August 2026
- MeitY, DPDP Rules, 2025 — G.S.R. (General Statutory Rules) 846(E), Accessed on: 4 August 2026
- LangGraph Documentation, https://langchain-ai.github.io/langgraph/, Accessed on: 25 July 2026
- RAGAS (Retrieval Augmented Generation Assessment) Docs, https://docs.ragas.io/, Accessed on: 4 August 2026